

BiLo-NSGA: Budget-Aware Project-Level Local Moves with Accepted-Move Logging for Power-Grid Portfolio Optimization

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Abstract

Utility review boards must select grid investment portfolios under hard annual budgets, yet generic evolutionary operators do not express changes in review terms. BiLo-NSGA embeds affordable insertion, atomic delete–insert substitution, a heuristic group-label bonus, and feasibility recovery within non-dominated sorting. We compare a metered stage-local BiLo-NSGA implementation with a fully specified NSGA-II and Pareto Local Search on 120 public-proxy candidates, eight scenarios, and 30 common seeds. Under an exact 3200-unit budget that charges every population-objective evaluation or evaluated local proposal, mean feasible-front hypervolume is 0.16277 for BiLo-NSGA, 0.17213 for NSGA-II, and 0.11626 for local search. BiLo-NSGA loses four of eight Holm-corrected NSGA-II contrasts and wins none, while exceeding local search in all eight. At equal 0.20-s search deadlines, the NSGA-II comparison has three significant wins, two losses, and three unresolved contrasts. Every final population is budget-feasible. Bound, clipping, reference-point, and local-parameter checks preserve the NSGA-II–BiLo–local-search pooled ordering but change gap magnitudes; the joint proposal-cap setting $(D_+, D_-) = (2, 2)$ is descriptively 4.29% above the registered $(8, 4)$ setting. Thus, matched evidence supports a feasible project-vocabulary search framework and a bounded local-search advantage, not superiority over NSGA-II. Event summaries are not lineage or replay records.

Keywords: power grid investment planning; project portfolio selection; budget constraint; multi-objective evolutionary optimization; NSGA-II; local search; accepted-move logging

1. Introduction

Power systems face substantial capital requirements from renewable integration, storage integration, feeder automation, and reinforcement backlogs. NERC reliability reports document the consequences associated with delayed investment in specific asset classes. Utilities must choose among reinforcements, automation retrofits, storage, and renewable-support projects spanning many zones and feeders, while annual budgets fund only a subset. Project review therefore becomes a recurring selection problem with a hard budget and conflicting objectives (cost, reliability benefit, renewable accommodation, and execution risk). The benchmark also assigns projects a common zone- or feeder-group label, but does not model joint dependency benefits or dependency constraints.

Current review practice balances inspectable scoring against portfolio interaction. AHP/TOPSIS-style scoring exposes each ranking step but evaluates projects independently, omitting budget crowding-out and conditional marginal value. Multi-objective evolutionary algorithms (MOEAs) search portfolios directly, but generic bit-level variation does not

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express changes in project-review terms. BiLo-NSGA therefore tests whether budget-aware project-level moves can improve the benchmark objective while producing run-level counts and pool-local co-occurrence summaries of accepted search events.

BiLo-NSGA uses a custom non-dominated-sorting kernel with NSGA-II-style constrained environmental selection and changes the variation stage through project-level local search. The forward pass inserts an affordable project while budget slack remains. The backward-side operation is an atomic substitution: it tentatively removes a weak selected project, inserts the strongest affordable replacement, and accepts or rejects the pair as one move. A fixed 1.06 heuristic bonus favors an insertion whose group label is already present, and deterministic recovery repairs budget violations. Accepted insertions, accepted substitutions, and repair drops are appended to a transient run-level list. The released run rows retain only event count and final-front pool-position co-occurrence; they do not retain parent-child lineage, states sufficient for replay, or a recommendation path.

The local-operator evidence is asymmetric. In the earlier fixed-generation archive, removing forward insertion is harmful in dependency-constrained review, local-move explainability, and project-pool scalability under that archive's declared family, but the ablation has the higher pooled mean because other settings move nominally in its favor. Removing atomic substitution raises the pooled mean by 0.61%, but the contrast remains unresolved; replacing it with legacy standalone deletion raises the pooled mean by 0.22%, and that contrast is also unresolved. The new matched-effort study sharpens the boundary: BiLo-NSGA is below NSGA-II under the exact combined evaluation budget, while equal-time comparisons vary by scenario. Substitution is retained as a paired remove-insert event, not claimed as an accuracy mechanism or a validated reviewer aid.

A second obstacle is the scarcity of public grid investment-review records. We therefore construct a reproducible benchmark of 120 candidate projects in six archetypes. The candidates are derived deterministically from RTS-GMLC production-cost data, SimBench distribution-network data, and cached public NERC report metadata. Eight scenarios vary the budget (0.75x–1.20x nominal), candidate-pool composition, and group-label structure. The candidate-derivation pipeline is shared with the methodologically distinct TRACE-MOEA companion study [33]; Section 2.4 and the Data Availability statement disclose this relationship.

The contributions of this paper are:

1. **Budget-aware project-level local moves with accepted-move logging.** BiLo-NSGA defines forward insertion, atomic delete-insert substitution, a heuristic group-label bonus, and feasibility recovery within non-dominated sorting. The implementation records accepted insertions and substitutions plus repair drops in memory and reduces them to run-level count and set-overlap fields. The retained summary reports 3668 events per run and 99.6% final-front pool-position co-occurrence on average (Section 4; Table 7); these values do not constitute an audit trail, lineage, replay record, or explanation-quality measure.
2. **A shared public benchmark used to isolate the local-search question.** The 120-candidate generator and public source corpora are shared with the TRACE-MOEA companion study; this paper's hard-budget formulation, BiLo-NSGA implementation, scenarios, run records, selected portfolios, and comparisons are reported as paper-specific (Sections 2.4 and 3.4).
3. **Matched-evaluation and matched-time comparisons with explicit multiplicity.** A new immutable archive contains 1440 runs of the metered stage-local BiLo-NSGA, NSGA-II, and Pareto Local Search implementations: 720 at exactly 3200 combined population-objective/local-proposal units and 720 at a common 0.20-s search deadline. The primary 16-contrast family shows four significant BiLo-NSGA losses to NSGA-II

and eight wins over the bounded local search; the separately corrected time family is mixed (Section 6.8).

4. **Bound, clipping, reference-point, and local-rule sensitivity with bounded attribution.** Six hypervolume schemes and a seven-cell one-factor local scan retain adverse and null results. The pooled method ordering is stable across the schemes, but scale and gap magnitude are not; the joint (2,2) proposal-cap setting is descriptively stronger than the registered (8,4) setting, while penalty changes are null because evaluated proposals remain feasible. Earlier atomic-substitution contrasts and the public-record checks remain unresolved or descriptive (Sections 6.3–6.5 and 6.8).

The remainder of the paper is organized as follows. Section 2 reviews related work. Section 3 formalizes the problem and describes the public benchmark. Section 4 presents BiLo-NSGA. Section 5 details the experimental protocol. Section 6 reports results. Section 7 discusses the findings, Section 8 states limitations, and Section 9 concludes.

2. Related Work

Our work sits at the intersection of three research threads: budget-constrained combinatorial optimization and project portfolio selection, investment planning and project review practice in power grids, and local-search-enhanced multi-objective evolutionary algorithms (MOEAs). We review each thread in turn and then position BiLo-NSGA against them and against its companion method TRACE-MOEA.

2.1. Budget-Constrained Combinatorial Optimization and Project Portfolio Selection

Selecting items under a budget is the multi-objective knapsack setting used by Zitzler and Thiele [1] to compare Pareto-based schemes. Problem-aware search is important in this landscape. Jaskiewicz [2] shows that local search over scalarizing functions can improve on purely recombinative MOEAs, a result reviewed by Lust and Teghem [3] and later knapsack taxonomies [4]. Many-objective knapsack problems further challenge standard MOEAs [5].

Portfolio studies respond with tailored search. Doerner et al. [6] apply Pareto ant colony optimization, while Carazo et al. [7] model dependencies, schedules, and resource limits with explicit feasibility handling. Energy applications include coupled risk–benefit portfolios [8] and hybrid exact–heuristic investment planning [9]. The cited studies generally enforce budgets after variation through penalties or repair; the reviewed examples do not define the neighborhood itself as insertion under slack, deletion under violation, and substitution near budget parity.

2.2. Power Grid Investment Planning and Project Review Decision Methods

Grid investment research divides between expansion optimization and multi-criteria review. Optimization studies coordinate transmission and storage [10], value reliability through interruption cost [11], assess grid-side storage returns [12], and balance battery investment against operating benefit [13]. These formulations usually size a specific asset class from a small set of alternatives rather than select from more than one hundred heterogeneous projects.

Review practice commonly uses AHP [14] and TOPSIS [15]. Recent extensions add objective weighting and explainability [16], while site-selection reviews show growing use of multi-objective optimization [17]. MCDM exposes scoring steps but scores projects largely independently, so it misses budget crowding-out and dependencies. Simulation-based assessments of substations and commissioning delays [18] add network context but

do not solve the portfolio search. The open gap examined here is therefore a portfolio-aware method whose committed project-level search events are recorded [19].

2.3. Local-Search-Enhanced Multi-Objective Evolutionary Algorithms

Hybridizing evolutionary search with local refinement is a long-standing multi-objective strategy. Ishibuchi and Murata [20] apply weighted-sum local moves after genetic variation, while Knowles and Corne [21] show that a $(1 + 1)$ local search with a Pareto archive can approximate nondominated fronts competitively. NSGA-II [22], MOEA/D [23], and NSGA-III [24] rely primarily on selection and recombination; Neri and Cotta [25] survey the many memetic extensions built around them.

Power-system studies apply improved NSGA-II variants to distribution-network co-optimization [26], offshore-wind storage configuration [27], and resilient backbone-grid planning [28]. Enhanced swarm methods address storage planning [29] and transmission expansion [30]. In the cited studies, method changes concern initialization, operators, or stages, and the reported local moves, when present, remain numeric perturbations inherited from continuous benchmarks.

Portfolio-scheduling studies also combine NSGA-II with Pareto local search [31]. More recent methods tailor local refinement to cardinality and pre-assignment constraints [32]. The cited hybrids do not report moves as adding, removing, or substituting a named project. BiLo-NSGA records accepted move types and pool-local positions during a run, then releases only aggregate count and set-overlap fields; it does not reconstruct a final-portfolio lineage.

2.4. Relation to TRACE-MOEA and Gap Statement

The TRACE-MOEA companion study [33] addresses power-grid project review through preference-adaptive elitism, deterministic repair, and run-level event co-occurrence summaries. Its contribution acts at selection and ranking. BiLo-NSGA instead changes variation through forward insertion, atomic substitution, a heuristic group-label bonus, and feasibility recovery. The two studies share the candidate generator and the public NERC and MTEP16 source corpora. Their problem objectives, algorithm implementations, scenario definitions, run archives, selected portfolios, and reported comparisons are paper-specific.

The resulting gap is specific. The reviewed knapsack MOEAs generally handle budgets through penalties or post-variation repair. The reviewed grid MCDM studies expose scoring steps but remain portfolio-blind, and the reviewed memetic MOEAs express refinement numerically rather than as named project edits. BiLo-NSGA combines non-dominated sorting with forward insertion, atomic replacement, feasibility recovery, and accepted-move/repair counters. The claim concerns this reviewed intersection rather than universal superiority over memetic optimization or validated recommendation provenance.

3. Problem Formulation and Public Benchmark

3.1. Budget-Constrained Portfolio Review as Multi-Objective Selection

Let $\mathcal{C} = \{p_1, \dots, p_n\}$ be the candidate pool. Each project has cost c_i , reliability benefit r_i , renewable-accommodation benefit g_i , load-support value ℓ_i , compliance score a_i , evidence score e_i , and schedule and implementation risks combined as ρ_i . A categorical group label d_i identifies a shared zone or feeder. The label is used only by the heuristic move score in Section 4.4: the formulation contains no group-level benefit term and no requirement that group members be selected together. A review decision is a binary vector $x \in \{0, 1\}^n$, where $x_i = 1$ means project p_i is funded. The review board's problem is:

Table 1. Candidate pool composition.

Source	Public artifact used	Candidates	Archetypes (kind)
RTS-GMLC	bus/branch/generator source data, zone-level aggregates	72	transmission reinforcement; reliability automation; renewable support
SimBench	complete mixed network (1-complete_data-mixed-asset-sw), 16 highest-stress subnets	16	distribution (feeder) reinforcement; storage flexibility; protection automation
NERC / C2GES report cache	metadata of 40 cached public reliability documents (28 event reports)	attribute adjustment only	—
Total		120	6 kinds

$$\min_{x \in \{0,1\}^n} F(x) = \left(\sum_i c_i x_i, -\sum_i r_i x_i, -\sum_i g_i x_i, f_4(x) \right), \quad 178$$

where the fourth objective follows the implemented empty-portfolio convention: 179

$$f_4(x) = \begin{cases} \frac{\sum_i \rho_i x_i}{\sum_i x_i}, & \sum_i x_i > 0, \\ 1, & \sum_i x_i = 0. \end{cases} \quad 180$$

subject to the **hard budget constraint** 181

$$\sum_i c_i x_i \leq B, \quad 182$$

with normalized constraint violation 183

$$v_B(x) = \max\left\{0, \frac{\sum_i c_i x_i - B}{B}\right\}. \quad 184$$

The four minimization objectives are thus total cost, negated reliability benefit, negated renewable benefit, and mean per-project risk with an explicit empty-portfolio value of 1. The budget is not a soft preference: portfolios exceeding B are not fundable, which is why the benchmark evaluates only the feasible non-dominated front (Section 5.3) and why the method design treats budget slack and violation as first-class quantities that shape the search neighborhood (Section 4). 185–190

A portfolio x dominates x' when it is at least as good in all four objectives and better in at least one. The non-dominated front contains mutually non-dominating portfolios returned by a method. Hypervolume measures the objective-space volume dominated by that front relative to a fixed reference point, rewarding both convergence and diversity. 191–194

3.2. Benchmark Construction from Public Data 195

Because real utility investment-review records are institution-internal and unavailable for publication, we derive a candidate pool deterministically from three public sources (Table 1). The full derivation code is released; every rule below is inspectable. 196–198

RTS-GMLC zone-level aggregates of load, branch ratings, permanent-outage rates, generator capacity, and renewable share generate three candidate archetypes per zone. Analytic functions map these aggregates to costs and benefits; for example, reinforcement benefit scales with outage pressure, while renewable-support benefit scales with the gap between load and installed renewable capacity. From SimBench, the sixteen subnets with the highest combined load and line-length stress each yield feeder-reinforcement, storage-flexibility, and protection-automation candidates. Their attributes depend on subnet load, 199–205

Table 2. The eight review scenarios. Pool filters restrict candidate kinds; the evaluation never changes.

Experiment	Budget multiplier	Candidate pool
budget_constrained_selection	0.88x	full pool (120)
reliability_prioritized_review	1.00x	reliability-related kinds only
renewable_accommodation_review	1.00x	renewable/storage kinds only
dependency_constrained_review	1.00x	filter retaining candidates whose group label occurs at least twice; no co-selection constraint
local_move_explainability	1.00x	full pool
ranking_robustness	1.00x	full pool
budget_sensitivity	0.75x	full pool
project_pool_scalability	1.20x	full pool

line count, and the distributed-energy-resource gap. NERC report metadata then adjusts attributes at the project-kind level. Candidates in the same zone or feeder receive a common group label, producing 120 candidates in six kinds with a block structure; the label does not create an evaluated joint benefit.

We emphasize what this construction is and is not. It is a *reproducible, public, engineering-plausible proxy* for the review problem – every attribute traces to public grid statistics or public reliability-report metadata through published rules. It is not a set of expert-labeled review outcomes, and its cost coefficients are in synthetic cost units, not calibrated currency (Section 8).

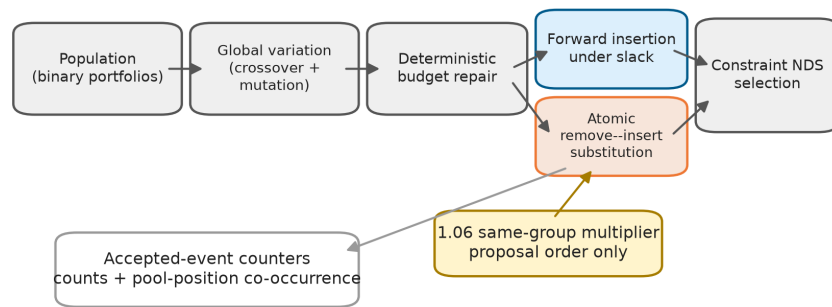
3.3. Review Scenarios

Eight experiments exercise the pool along three axes – budget envelope, pool composition, and group-label structure – while the evaluation itself (objectives, violation, and hypervolume computation) is identical across experiments (Table 2). The nominal budget is $B = 1020$ cost units.

Five experiments use the full candidate pool at budget multipliers 0.75x, 0.88x, 1.00x, 1.00x, and 1.20x. Their random streams are independent. The scenario-weight vector, ordered as reliability, renewable, load support, compliance, evidence, risk, and cost, is (0.26, 0.18, 0.20, 0.14, 0.12, 0.26, 0.38) by default. `budget_constrained_selection` and `budget_sensitivity` change the cost entry to 0.50; `dependency_constrained_review` changes reliability to 0.40 and risk to 0.36; `local_move_explainability` changes cost to 0.44 and risk to 0.32; and `renewable_accommodation_review` changes renewable to 0.42. The other scenarios retain the default vector, including `reliability_prioritized_review`. These unnormalized weights enter Greedy BCR, AHP-TOPSIS, and WeightedRankingOnly, but do not enter BiLo-NSGA’s local acceptance rule, the four objectives, or hypervolume. Because budgets, pools, weights, and random streams are not all held fixed, Section 6.2 treats the budget-indexed results as a cross-scenario diagnostic rather than a controlled budget-only experiment.

3.4. Shared Public Benchmark Statement

The Section 3.2 candidate pipeline is shared with the TRACE-MOEA companion study [33]. Both studies also use the same public NERC corpus and MTEP16 source records for separately executed consistency checks. Problem objectives, method implementations, scenarios, run records, selected portfolios, and reported comparisons are specific to each paper.



Evidence boundary: forward insertion resolves only in three legacy scenarios; atomic-substitution effects remain unresolved.

Figure 1. BiLo-NSGA architecture. The feedback arrow denotes the next generation. “Dependency-aware” in the diagram denotes only the 1.06 group-label proposal bonus, not a dependency objective or constraint. The lower elements show accepted-event counters as outputs and the 1.06 group multiplier that affects proposal ordering; neither enters an optimization objective.

4. BiLo-NSGA

Figure 1 places two project-vocabulary moves between global offspring generation and deterministic feasibility recovery. Forward insertion spends useful slack. Atomic substitution removes one weak selection and evaluates one affordable replacement before either change can be committed. A heuristic group-label bonus changes only the proposal order. The candidates then compete under constrained non-dominated sorting. Forward insertion produces resolved positive contrasts only in dependency-constrained review, local-move explainability, and project-pool scalability under the declared legacy family; substitution emits a paired remove–insert event without a demonstrated hypervolume gain.

BiLo-NSGA is a memetic multi-objective algorithm: a custom non-dominated-sorting kernel provides global search and Pareto pressure, and project-vocabulary local moves provide budget-aware intensification. Accepted local moves and repair drops are counted separately from the optimization objectives. Each component below is individually switchable, enabling the one-switch and legacy-rule comparisons in Section 6.3.

Formal definitions. The equations below specify the legacy implementation used for the broad archive; Section 5.4 discloses the metered stage-local differences used in the matched experiment. Let $x \in \{0, 1\}^n$ and define remaining budget and the same dimensionless violation used by evaluation and constraint dominance as

$$s_B(x) = B - \sum_{j=1}^n c_j x_j, \quad v_B(x) = \max\left(0, \frac{-s_B(x)}{B}\right).$$

At generation t , let P_t be the 40 parents and Y_t the 40 post-mutation children *before* repair or local search. The implementation freezes generation-local bounds

$$L_q^{(t)} = \min_{z \in P_t \cup Y_t} f_q(z), \quad H_q^{(t)} = \max_{z \in P_t \cup Y_t} f_q(z),$$

and uses

$$\hat{f}_q^{(t)}(x) = \frac{f_q(x) - L_q^{(t)}}{\max(H_q^{(t)} - L_q^{(t)}, 10^{-9})}.$$

The value is not clipped after a local move. The acceptance scalar used *inside local search only* is therefore

$$\Phi_t(x) = \sum_{q=1}^Q \hat{f}_q^{(t)}(x) + 10v_B(x). \quad 267$$

Thus the local penalty coefficient is $\lambda = 10$, every normalized objective has coefficient one, and the scenario weights of Section 3.3 do not enter Φ_t . These generation-local bounds are distinct from the fixed method-independent bounds used for hypervolume in Section 5.3.

The move-ranking numerator is a raw mixed-scale heuristic. Define

$$h_j = r_j + g_j + \ell_j + \frac{1}{2}(a_j + e_j), \quad R_j^+(x) = \frac{h_j}{\max(c_j, 1)} [1 + 0.06 \mathbb{I}(\exists k : x_k = 1, d_k = d_j)]. \quad 273$$

The four terms in h_j are not normalized across attributes and are not multiplied by scenario weights. The bracketed factor is only a proposal-ranking bonus. The forward proposal and acceptance rule are

$$j^+ = \arg \max_{j: x_j=0, c_j \leq s_B(x)} R_j^+(x), \quad x^+ = x + e_{j^+}, \quad 277$$

$$x \leftarrow x^+ \quad \text{iff} \quad \Phi_t(x^+) < \Phi_t(x); \quad \text{otherwise the forward pass stops.} \quad 278$$

For atomic substitution, first select a removal candidate

$$j^- = \arg \min_{j: x_j=1} \frac{h_j}{\max(c_j, 1)}, \quad x^- = x - e_{j^-}. \quad 280$$

The affordable replacement set excludes that project,

$$\mathcal{A}(x^-) = \{j : x_j^- = 0, j \neq j^-, c_j \leq s_B(x^-)\}, \quad 282$$

and $j^+ = \arg \max_{j \in \mathcal{A}(x^-)} R_j^+(x^-)$. The pair is atomic:

$$x \leftarrow x - e_{j^-} + e_{j^+} \quad \text{iff} \quad \mathcal{A}(x^-) \neq \emptyset \wedge |x^-| \geq 2 \wedge \Phi_t(x - e_{j^-} + e_{j^+}) < \Phi_t(x). \quad 284$$

Otherwise both tentative edits are rolled back and the pass stops. Thus the proposed method never accepts a standalone backward deletion; that legacy rule is retained only as a named ablation.

Eligible-index arrays are generated in ascending pool-local order. NumPy `argmin/argmax` therefore resolves exact ties in repair and local proposal scores to the smallest eligible pool-local index. Environmental crowding and the scalar-ranking baselines use `argsort` without an explicit stable secondary key, so equal values follow the implementation's returned order rather than a declared scientific tie rule. Exact cross-version tie replay is not claimed.

Feasibility recovery applies the same deletion ranking without the improvement test until $v_B(x) = 0$. For local depths D_+ and D_- , a direct scan implementation costs

$$T_{\text{local}} = O((D_+ + D_-)nQ) \quad 295$$

per selected offspring, in addition to objective evaluation. The current implementation uses $D_+ = 8$, $D_- = 4$; this complexity expression does not imply that both directions produce equal empirical benefit.

4.1. Non-Dominated Sorting Kernel

The custom kernel initializes 40 portfolios independently. For portfolio i , it draws $\delta_i \sim \text{Uniform}(0.03, 0.15)$ and then draws each bit as $x_{ij} \sim \text{Bernoulli}(\delta_i)$; deterministic budget repair is applied immediately. In each of 40 generations, two parent indices for each child are sampled independently and uniformly with replacement from the 40 current rows. A per-bit $\text{Bernoulli}(0.5)$ mask performs uniform crossover, followed by bit-flip mutation at rate $1/n$. All 40 children are repaired, and the first 20 rows in generated array order undergo local search. This is a fixed positional convention, not fitness-based or random offspring selection.

Parents and children are then pooled. Constraint dominance ranks feasible solutions above infeasible ones, compares infeasible solutions by the normalized v_B , and applies Pareto dominance among feasible solutions. Non-dominated fronts enter in order, and crowding distance truncates the final front to a population of 40. The procedure is NSGA-II-style environmental selection, but parent choice is uniform rather than binary tournament selection; it is therefore described as a custom non-dominated-sorting kernel rather than a canonical NSGA-II implementation.

4.2. Forward Insertion

After mutation and repair, the first 20 offspring undergo a forward pass of depth at most eight. While slack remains, the affordable unselected project with the highest raw $h_j / \max(c_j, 1)$ score after the optional 1.06 group bonus is tentatively inserted. The insertion is retained only if it improves Φ_t , which uses the frozen generation-local bounds and $\lambda = 10$. The pass stops at the first rejection. This move converts budget slack left by recombination into a candidate improvement.

4.3. Atomic Backward–Forward Substitution

The substitution pass has depth at most four. It removes the selected project with the lowest benefit-to-cost ratio only provisionally, uses the released budget to choose the highest-scoring affordable replacement, and evaluates the resulting portfolio once. Acceptance commits both edits; rejection restores both. The pass stops at the first rejection and never proposes from a portfolio with fewer than two retained projects. This construction records replacement as one paired event rather than as unrelated insert and delete events. Section 6.3 shows that the paired representation does not yield a significant hypervolume change, and its usefulness to reviewers is not evaluated.

4.4. Heuristic Dependency-Group Bonus

When ranking insertion or replacement candidates, the raw benefit-to-cost score is multiplied by 1.06 if another selected project has the same group label. The multiplier is a heuristic proposal preference only: it does not add a joint benefit to $F(x)$, impose a co-selection or precedence constraint, change acceptance, or enter evaluation. The `dependency_constrained_review` label denotes a pool filter plus scenario weights, not an additional constraint, and the `LowDependencyDensity` stress variant changes group labels. The present design therefore provides no evidence of dependency synergy or benefit.

4.5. Feasibility Recovery

Every initial or offspring portfolio that violates the budget is repaired deterministically. The selected project with the lowest raw $h_j / \max(c_j, 1)$ score is removed until the portfolio is affordable. The group bonus and scenario weights do not enter repair. Because each step removes one selected project and the empty portfolio is feasible, repair terminates after

at most $\|x\|_0$ drops. Each drop increments the move counter and appends a `repair_drop` event, including generation 0 initialization repair.

4.6. Accepted-Move and Repair Counters

Every accepted `forward_insert` and `backward_substitute`, and every deterministic `repair_drop`, increments `local_move_count` and appends one in-memory event. Rejected proposals and attempted moves are neither counted nor logged. Events store a generation label and pool-local integer position; a substitution stores the removed and inserted positions. The list includes initialization repairs and events from offspring that may later be discarded. It has no child or parent identifier, before/after state, scalar value, acceptance delta, or final-portfolio link.

The run writer does not serialize that event list. It retains `trace_event_count = len(events)` and `decision_coverage`, defined as the fraction of pool-local positions occurring anywhere in the deduplicated final feasible front that also occur anywhere in the run's event-position set. Because tracing is enabled for BiLo-NSGA and each counted action creates one event, `local_move_count` and `trace_event_count` are identical in the retained rows. The derived table reports means of 3667.954 for both fields and 0.9964326 for position co-occurrence across 240 runs (Table 7). No counter enters an objective, constraint, local acceptance, or environmental selection. These fields measure event production and set co-occurrence, not attempted-move rate, causal attribution, chronology, audit completeness, explanation quality, lineage, or replay.

4.7. Algorithm Summary

BiLo-NSGA(pool, budget, seed):

```

for i = 1 .. 40:
    draw density delta_i ~ Uniform(0.03, 0.15)
    draw each bit Bernoulli(delta_i); repair to budget and count/log each drop
for gen = 1 .. 40:
    sample two parent indices per child uniformly with replacement
    offspring[1:40] <- uniform crossover (bit mask 0.5) + bit flip (rate 1/n)
    freeze objective min/max from parents + pre-repair offspring
    for i = 1 .. 40:
        repair offspring[i] to budget (count/log repair_drop)
        if i <= 20:
            forward pass (depth <= 8): rank by raw BCR x heuristic group bonus,
                accept only strict Phi_t improvement; stop at rejection
            substitution pass (depth <= 4): tentatively remove lowest raw BCR,
                insert best affordable replacement, accept pair only on
                strict Phi_t improvement; stop at rejection
            count/log accepted forward_insert or backward_substitute only
        environmental selection: constraint-dominated NDS + crowding (top 40)
    return population; retain event count and final-front position co-occurrence

```

5. Experimental Setup

5.1. Methods Compared

Table 3 lists the eighteen methods: BiLo-NSGA, seven baselines spanning evolutionary, Pareto-local-search, and MCDM families, and ten ablations. The evolutionary baselines are pymoo implementations on the same binary problem, but their constraint paths differ

Table 3. Methods.

Method	Role	Description
BiLo-NSGA	proposed	custom constrained NDS + forward insertion + atomic substitution + heuristic group bonus + deterministic repair
NSGA-II	baseline	pymoo NSGA-II; normalized v_B passed as one inequality constraint
NSGA-III	baseline	pymoo NSGA-III with 35 Das–Dennis directions and population 40; normalized v_B constraint
MOEA/D	baseline	pymoo MOEA/D with 35 directions; no constraint interface, so $10^4 v_B$ is added to every objective
Greedy BCR	baseline	scenario-weighted benefit-cost order; add a project only while affordable
AHP-TOPSIS	baseline	scenario-weighted TOPSIS order; greedy affordable fill
Random Feasible	baseline	seeded random order; greedy affordable fill
Pareto Local Search	baseline	40 repaired random starts and feasible add/delete/swap moves; 1600-neighbor budget
Ablation-NoForwardSearch	ablation	forward insertion disabled
Ablation-NoBackwardSearch	ablation	atomic substitution disabled
Ablation-LegacyDeletion	ablation	atomic substitution replaced by standalone greedy deletion
Ablation-RandomMutationOnly	ablation	local search replaced by high-rate (3/n) random mutation
Ablation-NoDependencyMoves	ablation	heuristic group-label bonus disabled
Ablation-NoFeasibilityRecovery	ablation	deterministic repair disabled; custom constraint dominance retained
Ablation-WeightedRankingOnly	ablation	weighted ranking without evolution
Ablation-ShallowLocalSearch	ablation	local-search depth reduced to 2
Ablation-LowDependencyDensity	ablation	every third candidate receives an isolated group label
Ablation-LooseBudget	ablation	search at 1.2x budget, evaluated at the true budget

as disclosed below. Pareto Local Search uses add, delete, and swap neighborhoods with a fixed budget of 1600 evaluated neighbors per run.

Each operator ablation flips one switch relative to the full method. The last two are stress variants that perturb group labels and the search budget rather than removing an operator, and we read them as robustness probes.

5.2. Protocol

The rectangular archive contains **18 methods x 8 experiments x 30 invocations = 4320 rows**. Fourteen opponents are stochastic. AHP-TOPSIS, Greedy BCR, and Ablation-WeightedRankingOnly are deterministic; their repeated identical invocations do not increase the effective sample size beyond one per experiment. BiLo-NSGA, its evolutionary ablations, NSGA-II, and NSGA-III use population 40 for 40 generation labels. MOEA/D uses its 35 four-objective Das–Dennis directions as the effective population. The run archive does not retain function-evaluation counts, so identical evolutionary evaluation budgets are not claimed. Pareto Local Search receives a numerical ceiling of 1600 evaluated neighbors, matching the nominal 40×40 offspring ceiling of the population-40 methods but not their computational work; wall-clock runtime is reported separately.

Constraint handling is method-specific. The custom kernel uses normalized violation in constraint dominance and, for the full method, deterministic pre-search repair. Pymoo NSGA-II and NSGA-III receive v_B as an inequality constraint. Because the pymoo MOEA/D path has no constraint interface, it adds $10^4 v_B$ to every objective; this coefficient is unrelated to the local-search coefficient $\lambda = 10$. Greedy BCR, AHP-TOPSIS, and Random Feasible construct a feasible portfolio by skipping unaffordable projects, while Pareto Local Search repairs its starts, proposes only affordable additions or swaps, and filters its archive to feasible rows. Final hypervolume is computed from feasible non-dominated rows for every method.

The legacy protocol records the method list, scenarios, 30 seeds, nominal population 40, 40 generations, and evaluation description, but it does not capture every implementation choice needed for exact replay. Scenario weights enter only the three scalarizing rules named in Section 3.3 and never enter BiLo-NSGA or the evaluation metric. The supplementary artifact record documents the remaining replay limitations.

5.3. Evaluation Metric and Statistics

Portfolio quality is measured by the **standard hypervolume** of the feasible non-dominated front. Objective values are normalized with *fixed per-problem bounds* computed once per experiment from a seeded reference set (the empty portfolio, all single-project portfolios, and 2048 random feasible portfolios), and the reference point is 1.1 in every normalized dimension. These bounds are distinct from the generation-local, pre-repair parent-offspring bounds used only by Φ_t in Section 4. An empty feasible front scores zero. The metric contains no method-aware ingredient, and event counts and position co-occurrence are reported descriptively rather than entering any ranking.

Statistical comparisons use two-sided **Mann-Whitney U tests** between BiLo-NSGA and each of the fourteen stochastic opponents per experiment ($n = 30$ per group), with **Holm correction** within each stochastic family. We report rank-biserial effects and 5000-resample bootstrap confidence intervals for mean differences. The three deterministic rules are compared descriptively and receive no seed-level p-values.

The method-independent normalization for objective j is

$$\tilde{f}_j(x) = \min \left\{ 1, \max \left[0, \frac{f_j(x) - l_j}{u_j - l_j} \right] \right\},$$

where (l_j, u_j) comes from the frozen reference sample rather than any compared method. The scored set for method m , experiment e , and seed r is

$$\mathcal{P}_{mer} = \{x \in \mathcal{X}_{mer} : v_B(x) = 0, \nexists y \in \mathcal{X}_{mer} \text{ with } \tilde{F}(y) \prec \tilde{F}(x)\}.$$

Its standard four-objective hypervolume is

$$HV(\mathcal{P}_{mer}) = \lambda_4 \left(\bigcup_{x \in \mathcal{P}_{mer}} [\tilde{F}(x), z^{\text{ref}}] \right), \quad z^{\text{ref}} = (1.1, 1.1, 1.1, 1.1).$$

For two seed samples, the first-sample Mann-Whitney statistic is

$$U_1 = n_1 n_2 + \frac{n_1(n_1 + 1)}{2} - \sum_{i=1}^{n_1} R_i,$$

and the familywise adjustment over M opponents is

$$p_{(i)}^{\text{Holm}} = \max_{k \leq i} \min \left\{ 1, (M - k + 1) p_{(k)} \right\}.$$

For an experiment e , the relative effect reported against NSGA-II is

$$\Delta_e = 100 \frac{\overline{HV}_{e, \text{BiLo}} - \overline{HV}_{e, \text{NSGA-II}}}{\overline{HV}_{e, \text{NSGA-II}}} \%.$$

5.4. Immutable Matched-Effort Protocol

The fixed-generation archive above does not record function-evaluation counts and therefore cannot establish equal search effort. We consequently prespecified and executed a stage-local comparison of BiLo-NSGA, NSGA-II, and Pareto Local Search on the same

eight scenario pools, budgets, objectives, and 30 seed indices. The supplementary artifact record contains the complete design, implementation, per-run rows, and validation checks.

The primary protocol gives every method exactly **3200 search-evaluation units**, so $N_{\text{population}} + N_{\text{local}} = 3200$ for every primary run. A unit is either one initialization/offspring candidate scored on the four objectives or one evaluated local proposal. A local proposal is not double-counted as a population evaluation. Budget repair and archive ranking use already visible candidate attributes and do not consume an objective-evaluation unit. Final-front extraction and hypervolume are readouts after search and do not feed selection. This accounting equalizes scored candidates, not primitive CPU operations, generations, or neighborhood opportunities; the separate time protocol supplies the complementary computational estimand. BiLo-NSGA retains the disclosed low-density initialization, uniform crossover, $1/n$ bit mutation, deterministic repair, first-half local refinement, atomic substitution, and constraint non-dominated selection. Pareto Local Search uses 40 repaired starts and the disclosed add/delete/swap cycle. Its 40 starts plus 3160 proposals consume the same total as the other methods.

The metered BiLo-NSGA runner is not semantics-identical to the legacy broad-archive implementation. The legacy code freezes bounds on parents plus pre-repair offspring and uses a 10^{-9} denominator floor; the metered runner freezes bounds on parents plus the repaired child and uses 10^{-12} . The matched results therefore estimate this stage-local metered implementation, not an exact replay of the legacy implementation. The matched runner records evaluation, accepted-move, and repair counts but does not evaluate the legacy accepted-event log or its position-co-occurrence summary.

The recorded legacy optimizer could not be executed in the matched environment. The study therefore implements and labels a fully specified constrained NSGA-II: binary tournament by constraint rank and crowding, 0.9-probability two-point crossover, $1/n$ bit mutation, no repair, and constraint non-dominated environmental selection. The matched NSGA-II result applies to this disclosed implementation; equivalence to the earlier baseline is not claimed. Its final-feasibility rate of 1.0 is an empirical result across the retained runs, not a guarantee supplied by its variation operators. BiLo-NSGA and Pareto Local Search instead construct or repair candidates before retention.

The secondary protocol runs the same cells to a common **0.20-s search deadline**, checked before every candidate or proposal evaluation. Environmental ranking can produce a small recorded overshoot; the observed range is 0.200004–0.201374 s. Method order rotates with seed index. Search runtime, total invocation runtime, population-objective evaluations, local-proposal evaluations, final-population feasibility, feasible-front size, and feasible-front hypervolume are retained per run.

The primary multiplicity family comprises 16 paired contrasts: BiLo-NSGA versus two comparators in eight scenarios. Tests are exact two-sided paired sign tests over the 30 common seed indices, followed by one Holm correction. The 16 matched-time contrasts form a separate secondary family. Hypervolume and local-parameter sensitivity are descriptive and emit no p-values. The six hypervolume schemes cross the reported empirical bounds with clipping on/off, 25%-expanded empirical bounds, conservative analytic bounds, and reference points 1.05, 1.1, and 1.2. The local scan changes one of joint proposal-cap depth $\{2, 8, 16\}$, violation penalty $\{5, 10, 20\}$, or group bonus $\{1.00, 1.06, 1.12\}$ at a time, on all eight scenarios with ten common seeds per cell. Depth values 2, 8, and 16 set (D_+, D_-) to $(2, 2)$, $(8, 4)$, and $(16, 8)$, respectively; the scan does not estimate single-cap effects or interactions among depth, penalty, and bonus.

Table 4. Pooled leaderboard across eight experiments. Stochastic methods have 240 runs; deterministic-rule summaries derive from eight unique outputs. Final-front/event position co-occurrence is descriptive only.

Method	Role	Mean HV	Std	Mean runtime (s)	Position co-occurrence
Ablation-NoBackwardSearch	ablation	0.17294	0.00797	0.162	0.978
Ablation-NoForwardSearch	ablation	0.17257	0.00679	0.114	0.999
Ablation-ShallowLocalSearch	ablation	0.17236	0.00829	0.152	0.995
Ablation-LowDependencyDensity	ablation	0.17236	0.00849	0.218	0.996
Ablation-LegacyDeletion	ablation	0.17228	0.00844	0.193	0.984
Ablation-NoDependencyMoves	ablation	0.17198	0.00927	0.182	0.996
BiLo-NSGA	proposed	0.17190	0.00861	0.219	0.996
Ablation-RandomMutationOnly	ablation	0.17173	0.00701	0.050	0.999
NSGA-II	baseline	0.17000	0.00739	0.080	—
Ablation-NoFeasibilityRecovery	ablation	0.16811	0.01272	0.192	0.928
NSGA-III	baseline	0.16236	0.01331	0.100	—
Ablation-LooseBudget	ablation	0.16098	0.01351	0.240	0.996
AHP-TOPSIS	baseline	0.13875	0.00152	0.0004	—
Pareto Local Search	baseline	0.11636	0.01488	1.416	—
Random Feasible	baseline	0.07026	0.02089	0.0003	—
Greedy BCR	baseline	0.04098	0.00767	0.0003	—
Ablation-WeightedRankingOnly	ablation	0.03559	0.00892	0.0003	—
MOEA/D	baseline	0.02529	0.01404	0.304	—

6. Results

6.1. Legacy Fixed-Generation Comparison

Table 4 reports the previously archived fixed-generation summaries over eight experiments. This archive remains necessary for the broad baseline and ablation record, but its methods do not have verified equal objective-plus-proposal budgets. It is therefore historical context; Section 6.8 is the primary matched-effort comparison. Figure 2 shows the original distribution view, and Section 6.7 isolates the direct local-search control. Repeated deterministic-rule rows are summarized descriptively and are not treated as independent observations.

In this legacy archive, BiLo-NSGA attains pooled mean hypervolume 0.17190 (standard deviation 0.00861), 1.12% above the recorded NSGA-II (0.17000) while requiring 2.74 times its mean runtime. This is an unmatched quality–compute tradeoff, not an effort-controlled advantage. BiLo-NSGA is also 5.87% above NSGA-III and 23.9% above AHP-TOPSIS. Across five stochastic baselines and eight scenarios, it has 37 positive mean differences in 40 comparisons; 36 are Holm-significant wins and none is a significant loss. The new 3200-unit comparison reverses the pooled NSGA-II ordering (Section 6.8), so the manuscript’s headline conclusion follows the matched result.

Table 5 breaks out the comparison against NSGA-II, the only competitive baseline.

In the legacy archive, the 0.75x and renewable-filtered comparisons are unresolved nominal losses and the 0.88x mean gain also fails multiplicity correction. Runtime is 0.219 s versus 0.080 s for NSGA-II, a 2.74-fold cost, while Pareto Local Search uses 1.416 s. Because neither objective-call nor proposal counts were retained, this table cannot identify whether the 1.12% pooled difference comes from search logic or unequal work. The legacy pair is therefore a quality–compute tradeoff, and the unresolved fairness issue motivates Section 6.8.

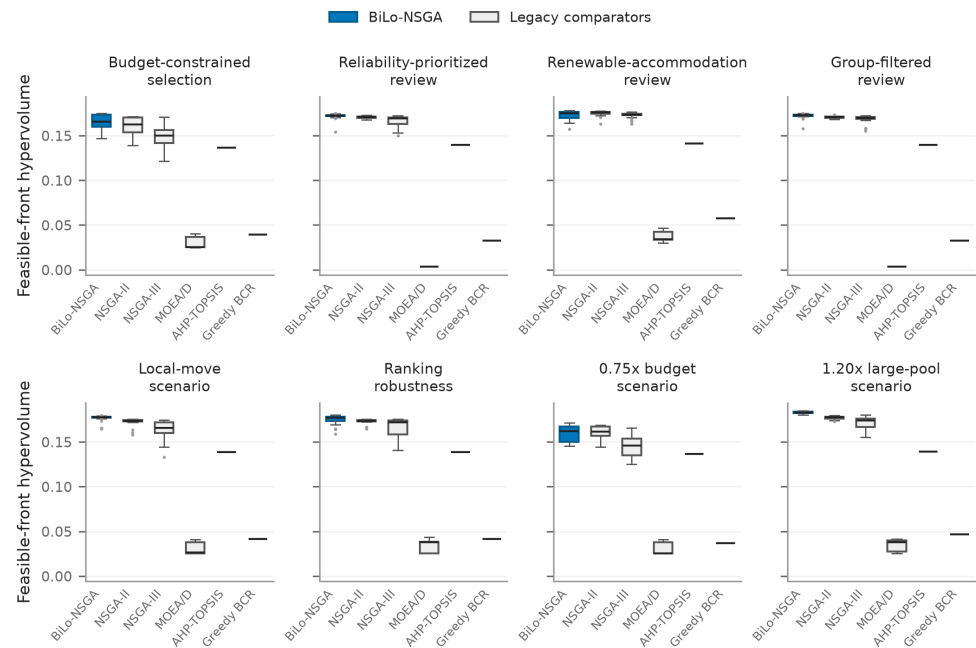


Figure 2. Hypervolume distributions across eight review scenarios (30 seeds per box for stochastic methods). Deterministic rules are point estimates rather than inferential samples. Pareto Local Search is reported in Table 4 and Figure 9.

6.2. Budget-Indexed Cross-Scenario Behavior

Five experiments share the full 120-candidate pool and span budget multipliers 0.75x, 0.88x, 1.00x, and 1.20x of the nominal 1020 cost units. Some labels also change scalar weights and all use independent random streams. Figure 3 is therefore a budget-indexed cross-scenario comparison, not a controlled budget-only scan.

Mean hypervolume rises with the budget index for the principal methods. BiLo-NSGA's margin over NSGA-II ranges from -0.64% at 0.75x and +2.17% at 0.88x to +3.30% in the 1.20x large-pool-labeled scenario. Because weights and seed streams are not held constant, this pattern cannot be attributed to budget headroom or forward insertion alone. The Ablation-LooseBudget stress rule, which searches at 1.2x and evaluates at the true constraint, is 6.35% below the full method when pooled on this benchmark. This supports enforcing the declared budget during search in the tested setup, not a general monotone budget-regime claim.

6.3. Ablation Study: Scenario-Dependent Forward Effects and Unresolved Substitution

Figure 4 compares the full method with all ten ablations (pooled over 8 experiments x 30 seeds).

Removing evolution entirely reduces pooled hypervolume by 79.3%. Disabling feasibility recovery costs 2.21%, increases the standard deviation from 0.00861 to 0.01272, and lowers final-front/event position co-occurrence from 0.996 to 0.928. By contrast, several local-operator ablations have pooled means above the full model. This prevents pooled ranking from being used as component attribution and makes the scenario-level tests essential.

Forward insertion is the only local operator with resolved positive cells in the legacy archive, but it is not universally beneficial. Ablation-NoForwardSearch has pooled mean 0.17257, 0.39% above the full method. Under that archive's declared within-scenario family, the full method exceeds it only in dependency-constrained review, local-move explainability, and project-pool scalability. The other five scenario comparisons remain

Table 5. BiLo-NSGA versus NSGA-II per experiment (30 seeds). Confidence intervals are pointwise, multiplicity-unadjusted 5000-resample bootstrap intervals for the mean difference; r_{rb} is rank-biserial. Holm-adjusted rank-test p-values determine significance across fourteen stochastic opponents within each experiment.

Experiment	BiLo-NSGA	NSGA-II	Mean difference (95% CI)	r_{rb}	Holm p
budget_constrained_selection (0.88x)	0.16434	0.16084	0.00349 [-0.00098, 0.00801]	0.209	1.000
budget_sensitivity (0.75x)	0.15965	0.16067	-0.00102 [-0.00506, 0.00291]	-0.022	1.000
dependency_constrained_review	0.17208	0.17043	0.00166 [0.00038, 0.00264]	0.751	5.44e-6
local_move_explainability	0.17369	0.17239	0.00429 [0.00243, 0.00626]	0.844	1.81e-7
project_pool_scalability (1.20x)	0.18288	0.17703	0.00584 [0.00514, 0.00657]	1.000	3.94e-10
ranking_robustness	0.17525	0.17328	0.00197 [-0.00008, 0.00378]	0.518	0.00529
reliability_prioritized_review	0.17169	0.17047	0.00122 [-0.00032, 0.00234]	0.607	0.000505
renewable_accommodation	0.17262	0.17490	-0.00228 [-0.00439, -0.00032]	-0.191	1.000

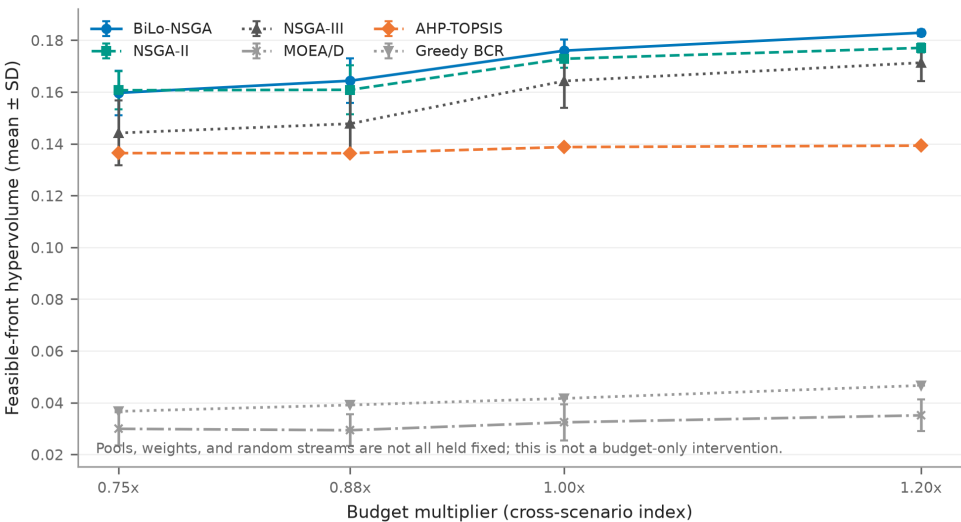


Figure 3. Mean hypervolume (+- std across seeds) as a function of the budget multiplier for the six main methods; the two nominal-budget experiments are pooled at 1.00x.

unresolved. The evidence therefore identifies three scenario-dependent forward-side cells, not a pooled or universal benefit from insertion.

Atomic substitution has no demonstrated accuracy gain. Ablation-NoBackwardSearch reaches 0.17294, 0.61% above the full method, and none of the eight differences is significant. The legacy standalone-deletion rule reaches 0.17228, 0.22% above the full method, and is likewise inseparable in all eight scenarios. Feasibility recovery already removes weak projects after violations, leaving few replacements that improve the scalar acceptance score. The atomic rule encodes removal and insertion as one committed event; the experiment does not demonstrate accuracy gain or reviewer utility.

The substitution operator records a single remove–insert pair and cannot leave a deletion-only intermediate portfolio. Because the no-forward control has the higher pooled mean while the full method resolves only in dependency-constrained review, local-move explainability, and project-pool scalability, the evidence does not select a preferred configuration. Alternative acceptance based on dominance or bounded look-ahead is a redesign target, not part of the present claim.

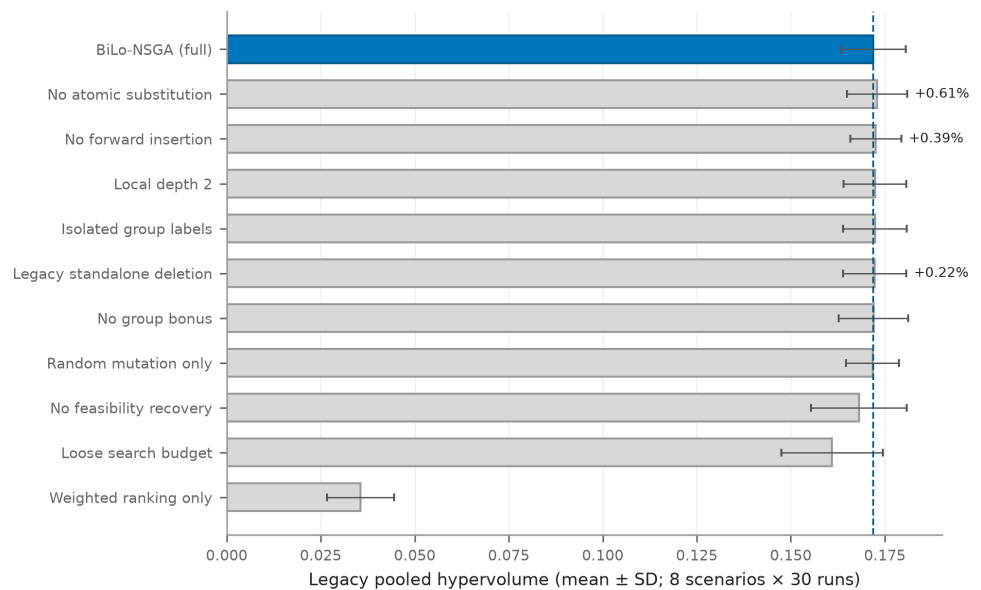


Figure 4. Mean hypervolume ($\pm$ std) of BiLo-NSGA and ten ablations or stress controls, pooled over all experiments and seeds. Formal decisions remain scenario-specific and Holm-adjusted.

Figure 5 separates event production from optimization quality. The full atomic-substitution configuration has higher final-front/event position co-occurrence than the no-backward variant but does not improve hypervolume at the available statistical resolution. Disabling forward search changes both event composition and scenario-level quality. These diagnostics do not imply that a larger event count or greater set overlap is intrinsically better.

The two stress variants bound other claims. Reassigning every third candidate to an isolated group label changes the pooled mean by less than 0.3%, so no group-density or dependency benefit is claimed. The LooseBudget variant's 6.35% deficit was discussed in Section 6.2.

6.4. External Consistency: NERC Rule-Based Backtest

Hypervolume measures optimization quality on the benchmark objectives; it cannot show whether selected portfolios point at real risk. We therefore backtest selections against cached NERC reliability reports using a rule designed to reduce construct circularity. Each candidate receives a NERC-topic kind weight multiplied by a stress percentile computed from raw RTS/SimBench attributes before the NERC-based adjustment of Section 3.2. The priority-capture ratio measures oversampling of high-priority candidates relative to a uniform draw, with 1.0 denoting parity. Kendall's τ measures rank alignment between selection frequency and priority. Figure 6 shows both review scenarios.

BiLo-NSGA oversamples documented-risk candidates, with capture ratios of 1.617 in budget-constrained selection and 1.301 in reliability-prioritized review. These values are above NSGA-II in the first setting (1.575) and below it in the second (1.403). Kendall τ values of 0.109 and 0.073 are not significant. AHP-TOPSIS has the highest nominal alignment because its reliability-type attributes overlap the rule. The backtest therefore indicates **descriptive external consistency only, not method superiority or ground-truth validity**. The stress-percentile component is independently derived, but kind-level weights retain construct overlap with pool generation.

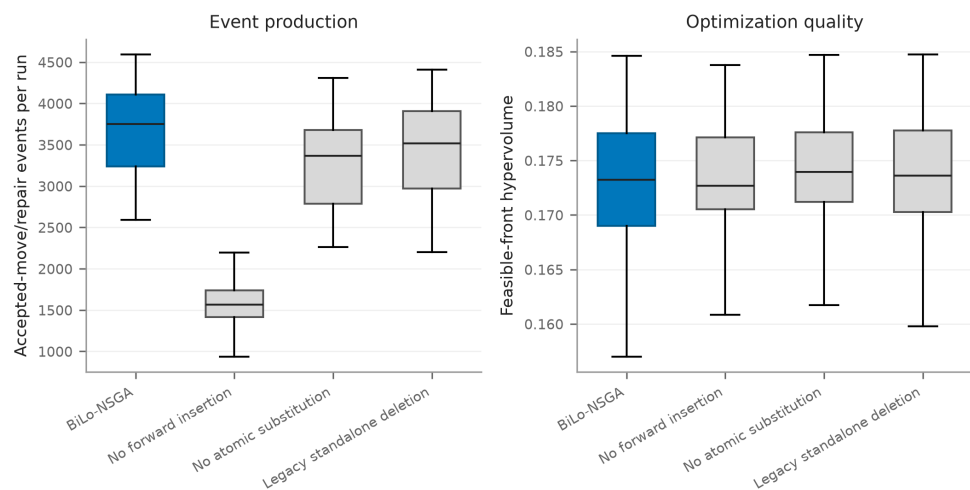


Figure 5. Local-search diagnostics pooled over eight experiments and 30 seeds (240 runs per configuration). (a) Accepted-local-move and repair-event counts per run; (b) seed-level mean-front hypervolume. Boxes are descriptive aggregates. Formal component decisions use per-experiment Holm-adjusted tests.

6.5. Historical Outcome Consistency: MISO MTEP16 Backtest

The NERC backtest of Section 6.4 checks consistency against documented risk patterns from the same public corpus used in benchmark construction. A more independent check compares selections with **observed transmission-project outcomes** from the MISO MTEP16 Appendix A/B inventory. Outcome snapshots were not used in fitting. However, decision-time appendix status is a prognostic feature, creating construct overlap with the broad outcome definition.

6.5.1. Data and Protocol

The MTEP16 dataset comprises 1218 projects with their 2016 cost estimates, project types, voltage levels, mileage, appendix status, and record dates. Quarterly Appendix A status snapshots from 2016-12 through 2018-01, together with the 2026 MISO portal in-service and active-project lists, produce three outcome classes:

- **Built** (n = 924): project ID appears in the 2026 in-service list or shows ‘In Service’ in a snapshot.
- **Withdrawn** (n = 19): explicitly withdrawn, never in service.
- **Deferred** (n = 39): still active in the 2026 Appendix A, excluded from capture metrics.
- **Unresolved** (n = 236): no trace in any 2026 list and never withdrawn; used only in the *broad* negative definition.

For portability, each method selects compromise portfolios from 10 seeded runs, preserving 2016 cost estimates after one global scaling factor sets the flagship budget to 5% of total pool cost. Point-biserial correlation and Mann–Whitney U compare selection frequency with broad or strict outcome labels; capture ratios relative to uniform selection are effect-size readouts. These are nominal project-level diagnostics. They do not preserve dependence among projects selected in the same portfolio, and no constrained-permutation or multiple-comparison family was preregistered. Two variants are shown: budget-constrained selection (1097 positive-cost projects) and reliability-prioritized review (1062 projects).

6.5.2. Results

Table 6 summarizes the key comparisons.

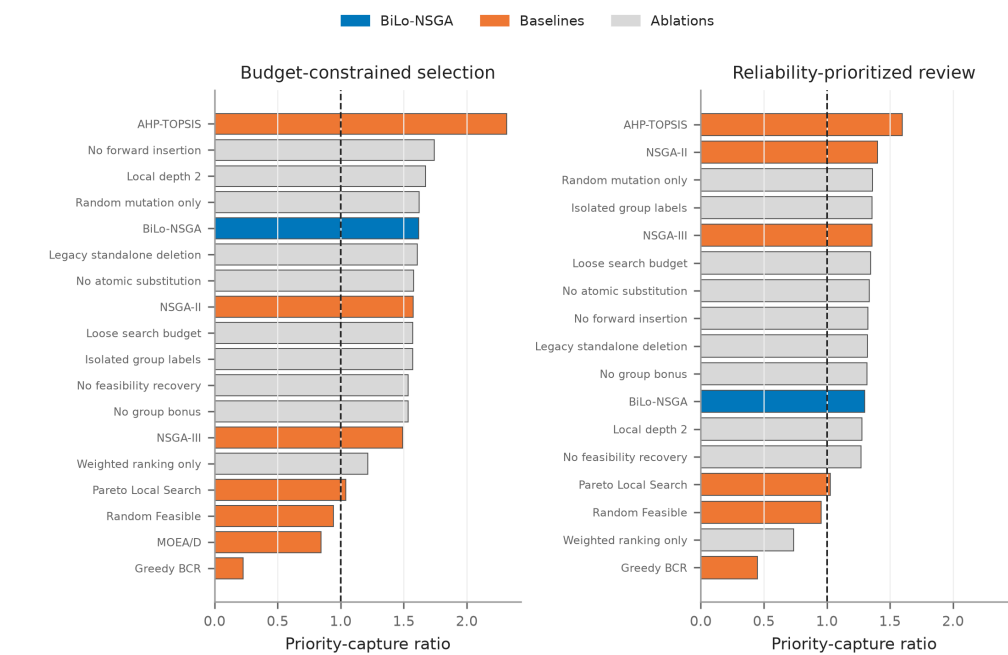


Figure 6. Priority-capture ratios in the NERC rule backtest (dashed line = parity with random selection). MOEA/D is absent from the reliability panel because it returned no feasible portfolio.

Table 6. MTEP16 historical backtest in the budget-constrained experiment. P-values are nominal project-level diagnostics, not confirmatory portfolio-level tests.

Method	Role	Capture (broad)	r_pb (broad)	p (broad)	MW p (broad)	Portfolio size
AHP-TOPSIS	baseline	1.0995	0.128	0.00003	0.00003	320.0
Ablation-NoFeasibilityRecovery	ablation	1.0877	0.091	0.0031	0.0084	200.8
Ablation-LowDependencyDensity	ablation	1.0716	0.089	0.0037	0.0173	247.0
BiLo-NSGA	proposed	1.0715	0.088	0.0042	0.0147	243.0
Pareto Local Search	baseline	1.0343	0.049	0.109	0.067	166.2
NSGA-III	baseline	1.0283	0.050	0.101	0.090	110.1
Greedy BCR	baseline	1.0267	0.057	0.062	0.062	585.0
NSGA-II	baseline	1.0118	0.025	0.425	0.264	124.7
Random Feasible	baseline	0.9893	-0.019	0.534	0.686	93.0

BiLo-NSGA’s broad capture is 1.071, or 7.1% above the numerical uniform-draw reference. The raw diagnostics are point-biserial $r = 0.088$ ($p = 0.0042$) and Mann–Whitney $p = 0.0147$. It exceeds the tested evolutionary and local-search baselines on broad capture, including Pareto Local Search at 1.034 and NSGA-II at 1.012, but AHP-TOPSIS is higher at 1.100. NoFeasibilityRecovery reaches 1.088 and LowDependencyDensity 1.072. The reliability-prioritized variant is directionally similar (capture 1.064, $r = 0.078$, raw $p = 0.0128$). These descriptive results do not identify a local operator or establish portfolio-level significance.

AHP-TOPSIS achieves the highest nominal broad capture (1.100) because it directly optimizes evidence/compliance features that overlap Appendix-status information. Within the strict built-versus-withdrawn subset, BiLo-NSGA’s capture is 1.009; MOEA/D and NSGA-II are nominally higher at 1.011 and 1.010. We therefore present the rerun as a **descriptive external-consistency result**. Strict labels remain underpowered, alternative ablations can align more strongly, and no superiority claim is made.

6.5.3. Outcome-Definition and Dependence Limits

We state the caveats that limit the strength of this backtest.

- **Base-rate ceiling.** MTEP16 Appendix A projects were overwhelmingly built (~98% within the strict subset). MTEP board approval itself is a strong filter; this backtest measures alignment *within* an already-approved plan, not the value of the review filter itself.
- **Low strict power.** With only 19 explicit withdrawals, the strict-label Mann-Whitney and point-biserial tests are low-powered, and a null result is expected even for a well-aligned method. The broad view has more negative instances (withdrawn + unresolved) but the unresolved class may include projects rebuilt under new MTEP IDs.
- **Type-distribution shift.** The real MTEP16 pool is dominated by reliability, asset-condition, and distribution projects; renewable and storage projects are nearly absent. The pipeline's renewable objective is therefore close to inert in this backtest, and renewable-related claims receive no support from this rung.
- **Appendix status as feature.** The `appendix_status` attribute (A / B-to-A / B) is used as an evidence feature. This is decision-time information (2016 board approval state), not an outcome, but it correlates with broad outcomes. Broad capture partly rewards methods that weight evidence/compliance features, which is why AHP-TOPSIS excels; strict capture (within board-approved Appendix A projects) is the cleaner discriminator.
- **Protocol differences.** Unlike the main experiments (30 seeds, hypervolume evaluation), the MTEP backtest uses 10 seeded runs and compromise-portfolio selection to align with real portfolio sizes. Point-biserial and Mann-Whitney are the primary readouts; capture ratios are effect-size indicators given the ceiling constraints.

6.6. Search-Effort and Outcome-Definition Diagnostics

For the legacy fixed-generation archive, computation and event-summary production are shown beside hypervolume. BiLo-NSGA averages 0.17190 hypervolume, 0.2188 s, 3668 accepted-move/repair events counted in memory, and 0.996 final-front/event position co-occurrence. NSGA-II averages 0.17000 at 0.0798 s and has no instrumented event list. The historical 1.12% quality margin at 2.74 times the runtime is an unmatched quality–compute tradeoff; Section 6.8 supplies the matched estimands. The released count and overlap fields are additional outputs, not evidence that a final recommendation can be reconstructed.

Table 7 makes the operator evidence operational. Removing atomic substitution reduces runtime by about 26% and nominally raises hypervolume by 0.61%, with all eight scenario contrasts unresolved under the declared comparison family. The contrast with standalone deletion also remains unresolved in all eight scenarios. Removing forward search lowers runtime and raises the pooled mean, but the full method resolves in dependency-constrained review, local-move explainability, and project-pool scalability only. These results describe a quality–runtime–event-production trade-off; the count and overlap fields do not establish which configuration a review process should use.

Figure 8 exposes outcome-definition sensitivity. BiLo-NSGA's broad capture is 1.0715 and 1.0642 across the two scenarios, whereas strict capture is 1.0093 and 1.0042. Raw broad correlations are nonzero under conventional project-level tests, but broad negatives include unresolved projects and portfolio dependence is unmodeled. The strict comparison is less ambiguous, yet contains only 19 withdrawals and has little power.

These legacy diagnostics assemble two views: fixed-generation optimization and event production, and outcome consistency under competing label definitions. They do not establish gain per equal unit of search. The retained outputs are not move histories, atomic substitution is not supported as an accuracy mechanism, and the historical backtest remains descriptive.

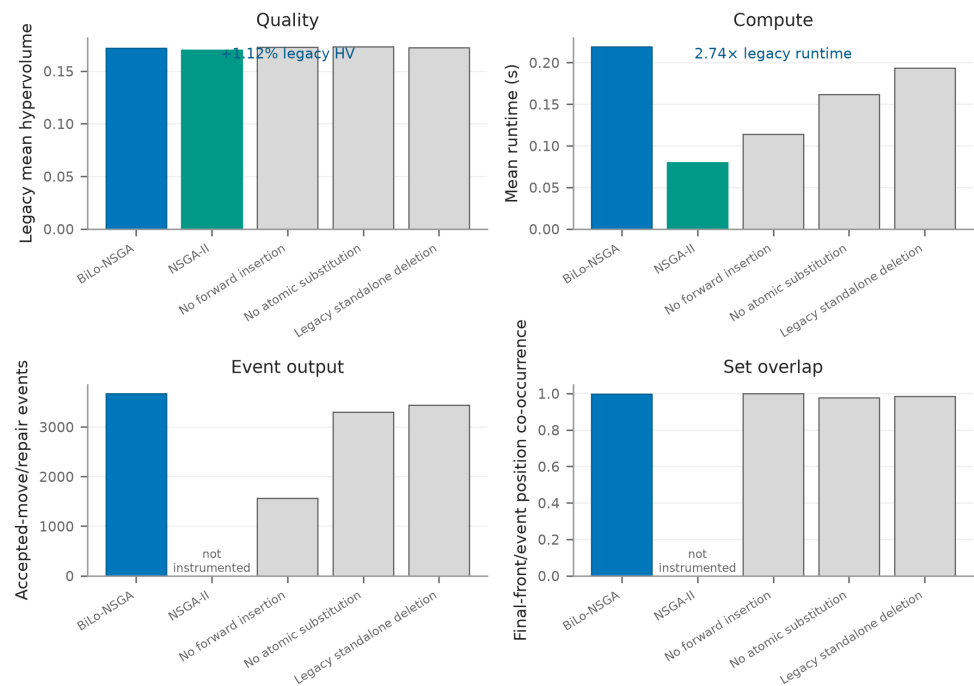


Figure 7. Pooled search-and-event diagnostics over eight experiments and 30 seeds. The figure's historical "trace" label denotes run-level count and position co-occurrence only. Each panel retains its native unit; no composite score is formed from optimization quality and event volume.

6.7. Direct Local-Search and Substitution Controls

In the legacy archive, Pareto Local Search supplies a direct non-evolutionary neighborhood-search baseline with add, delete, and swap moves under 1600 evaluated neighbors per run. Its pooled hypervolume is 0.11636, compared with 0.17190 for BiLo-NSGA, and the proposed method is significantly higher in all eight scenarios after Holm correction. The numerical ceilings do not capture BiLo-NSGA's local proposals, so this is not the exact combined-budget result. Section 6.8 reproduces the eight wins under both exact total units and equal time. The finding applies to the disclosed PLS configuration rather than to Pareto local search as a family.

Figure 9 places that baseline beside the three operator controls. NoBackwardSearch (0.17294) and LegacyDeletion (0.17228) both have higher pooled means than the atomic full method, and all eight scenario-level comparisons remain unresolved for each substitution control. NoForwardSearch (0.17257) also has a higher pooled mean, while the full method resolves only in dependency-constrained review, local-move explainability, and project-pool scalability. The figure therefore closes a comparator gap without creating a bidirectional-gain claim.

6.8. Matched-Evaluation, Matched-Time, and Sensitivity Results

6.8.1. Exact Combined Evaluation Budget

Table 8 reports the new primary comparison. All 720 method–scenario–seed runs consume exactly 3200 units, all final populations are budget-feasible, and every feasible front is nonempty. BiLo-NSGA spends on average 918.9 units on initialization/offspring scoring and 2281.1 on local proposals. NSGA-II spends all 3200 on population candidates; Pareto Local Search spends 40 on starts and 3160 on proposals. Thus, the total is identical while the allocation is algorithm-specific, as required by their search structures.

The pooled BiLo-NSGA mean is 5.44% below NSGA-II and 40.00% above Pareto Local Search. Against NSGA-II, every scenario mean is negative; the budget-constrained,

Table 7. Selected pooled quality–effort readouts.

Method	Mean HV	Runtime (s)	Accepted-move/repair event count	Position co-occurrence	Evidence implication
BiLo-NSGA	0.17190	0.2188	3668.0	0.996	1.12% legacy HV margin at 2.74x runtime versus NSGA-II nominally higher HV; all eight substitution contrasts unresolved all eight standalone-deletion contrasts unresolved higher pooled HV; full resolves only dependency-constrained review, local-move explainability, and project-pool scalability strongest external evolutionary baseline very low cost, lower proxy-objective hypervolume
NoBackwardSearch	0.17294	0.1616	3295.2	0.978	
LegacyDeletion	0.17228	0.1933	3438.6	0.984	
NoForwardSearch	0.17257	0.1139	1563.6	0.999	
NSGA-II	0.17000	0.0798	0	—	
AHP-TOPSIS	0.13875	0.0004	0	—	

Table 8. Mean feasible-front hypervolume under exactly 3200 combined evaluation units (30 common seeds per cell). A dagger marks a Holm-significant BiLo-NSGA contrast in the declared 16-test primary family; all four daggers against NSGA-II are losses and all eight against PLS are wins.

Scenario	BiLo-NSGA	NSGA-II	Pareto Local Search
budget_constrained_selection	0.14899	0.16987 [†]	0.10517 [†]
reliability_prioritized_review	0.16634	0.17006	0.11574 [†]
renewable_accommodation_review	0.16545	0.17569 [†]	0.12537 [†]
dependency_constrained_review	0.16840	0.17024	0.11595 [†]
local_move_explainability	0.16357	0.17419	0.12190 [†]
ranking_robustness	0.16755	0.17389 [†]	0.11507 [†]
budget_sensitivity	0.14404	0.16587 [†]	0.10016 [†]
project_pool_scalability	0.17578	0.17723	0.13071 [†]
Pooled descriptive mean	0.16277	0.17213	0.11626

renewable, ranking-robustness, and 0.75x-budget contrasts survive Holm correction, while the other four are unresolved. Against Pareto Local Search, all 30 paired differences are positive in every scenario and all eight contrasts survive correction. The result supports a bounded advantage over this PLS configuration but directly rejects an equal-evaluation NSGA-II superiority claim.

The runtime readout does not rescue an efficiency claim under this protocol. Mean search time is 0.1163 s for BiLo-NSGA, 0.1975 s for NSGA-II, and 0.2598 s for PLS on this host, but algorithms allocate the same total units to operations of different computational cost. Runtime is consequently paired with, not substituted for, the evaluation-budget estimand.

6.8.2. Equal Search Deadlines

At the common 0.20-s deadline, pooled means are 0.17098 for BiLo-NSGA, 0.17221 for NSGA-II, and 0.11593 for PLS. BiLo-NSGA receives more combined units on average (5601.1) than NSGA-II (3290.8) or PLS (2504.6) because its local proposal/scoring path is cheaper on this implementation. The NSGA-II contrast is heterogeneous: BiLo-NSGA has significant wins in reliability-prioritized, dependency-constrained, and project-pool-scalability settings; significant losses in budget-constrained selection and the 0.75x budget;

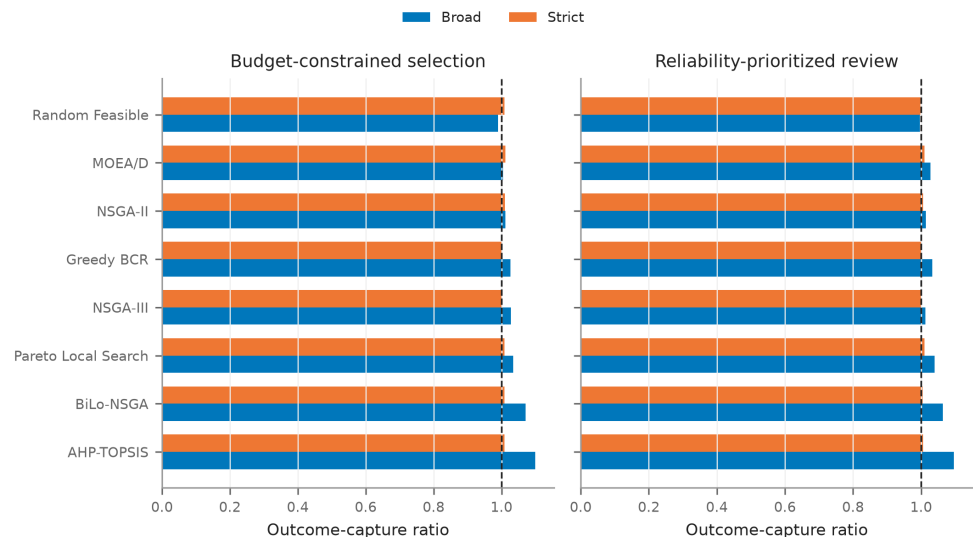


Figure 8. MTEP16 outcome-capture ratios for the proposed method and baselines. The dashed line denotes parity with uniform selection. Broad and strict labels are shown separately so that the high built-project base rate and unresolved-project assumption remain visible.

and unresolved renewable, local-move, and ranking-robustness contrasts. All eight PLS contrasts favor BiLo-NSGA. This separately corrected family supports scenario-dependent equal-time behavior, not an overall NSGA-II win or loss.

6.8.3. Hypervolume and Local-Rule Sensitivity

Across the 720 matched-evaluation fronts, the reported empirical bounds place 1121 of 28,061 front points (3.99%) outside $[0, 1]$ before clipping. Expanding those bounds by 25% reduces the count to 197 (0.70%); the conservative analytic bounds contain every point within the declared 10^{-12} tolerance. Pooled means change materially with the scale: the reported clipped values are 0.16277, 0.17213, and 0.11626 for BiLo-NSGA, NSGA-II, and PLS; reported bounds without clipping give 0.16547, 0.20209, and 0.11626; analytic bounds at reference 1.1 give 0.02584, 0.02854, and 0.02278. Analytic reference points 1.05 and 1.2 change the absolute volumes again. The pooled ordering NSGA-II–BiLo-NSGA–PLS is stable across all six schemes, but neither the numerical scale nor the gap magnitude is normalization-invariant.

The seven-cell local scan is descriptive and retains every adverse or null cell. The depth cells jointly set (D_+, D_-) to $(2, 2)$, $(8, 4)$, and $(16, 8)$. Relative to the registered $(8, 4)$ setting, $(2, 2)$ raises the scenario-balanced mean by 0.00691 (4.29%) and $(16, 8)$ by 0.00035 (0.22%). Changing the group-order bonus from 1.06 to 1.00 changes the mean by -0.00023 (-0.15%); 1.12 changes it by +0.00016 (+0.10%). Penalties 5, 10, and 20 produce identical rows because all locally evaluated proposals are feasible, so the violation term is zero. These results identify $(8, 4)$ as an unoptimized fixed choice and provide no measured accuracy benefit for the 1.06 bonus. Because the scan changes one factor at a time, it does not estimate interactions among proposal caps, penalty, and bonus. The results are not used to replace or retune the immutable primary run.

7. Discussion

Why the legacy operator effects are forward-side. Under deterministic repair, populations remain near the budget boundary. Forward insertion can exploit residual slack directly. Atomic substitution must find a removal, an affordable replacement, and a joint

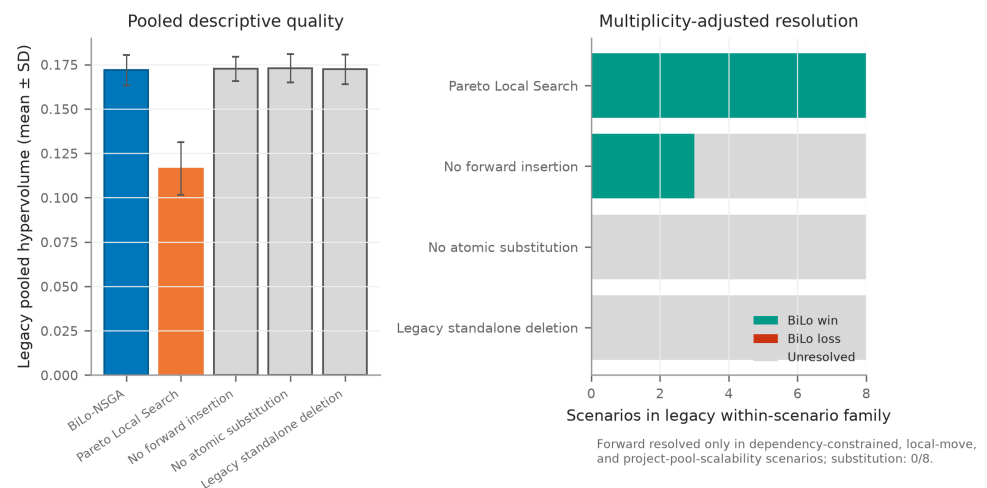


Figure 9. Pooled hypervolume and scenario-level significance counts for BiLo-NSGA, Pareto Local Search, NoForwardSearch, NoBackwardSearch, and LegacyDeletion. The direct PLS control is separated in all eight scenarios. Forward insertion resolves only in dependency-constrained review, local-move explainability, and project-pool scalability; both substitution-control families remain unresolved in all eight scenarios.

scalar improvement in one step, while repair has already removed many weak projects. This makes substitution opportunities rarer, but the explanation remains an inference from the earlier event and ablation patterns rather than a proved mechanism. The matched-evaluation loss to NSGA-II prevents these component contrasts from being promoted to an overall algorithmic advantage.

The operator effects depend on the scenario. The no-forward ablation has a higher pooled mean, yet the full method resolves only in dependency-constrained review, local-move explainability, and project-pool scalability; no corresponding resolved gain exists for atomic substitution. The full substitution rule is retained because it emits one paired replacement event, not because the study demonstrates reviewer utility. A look-ahead or dominance-based acceptance rule is the appropriate redesign target.

What the external-consistency checks add. The NERC rule check (Section 6.4) shows concentration on documented risk patterns but retains construct overlap with pool generation. The MTEP16 backtest (Section 6.5) uses independent historical outcomes and 2016-vintage features, but its project-level diagnostics ignore portfolio dependence and face a severe outcome base rate. Together, they test whether optimized portfolios immediately contradict two public-record views; they do not replace expert-labeled feasibility outcomes or calibrated costs.

Positioning against TRACE-MOEA. The TRACE-MOEA companion study [33] uses preference-adaptive elitism, deterministic repair, and run-level event co-occurrence summaries over the same public candidate pipeline. BiLo-NSGA instead studies project-vocabulary variation under a hard budget. Its central questions concern forward insertion, atomic substitution, and budget-indexed cross-scenario behavior. The methods share candidate generation and public-record infrastructure but not their core operators, configurations, executions, or analyses.

Practical implications. Equal evaluation budgets favor the disclosed NSGA-II in pooled hypervolume and produce four significant BiLo-NSGA losses, while equal time produces three BiLo-NSGA wins, two losses, and three unresolved contrasts. This is a quality-allocation trade-off: BiLo-NSGA's inexpensive local proposals permit more scored moves per second, but spending most of a fixed unit budget on those proposals is less effective than NSGA-II's population evaluations in this study. The joint (2,2)

proposal-cap setting is descriptively stronger than the registered (8,4) setting, so the committed configuration should not be recommended without a separately registered tuning or validation study. Pareto Local Search remains weaker under both matched protocols, but only as the bounded implementation tested here.

The 99.6% statistic is the set overlap between pool-local positions appearing anywhere in the returned fronts and anywhere in the run's event-position set. It is not per-portfolio coverage or explanation sufficiency. If single-criterion regulatory alignment is dominant, AHP-TOPSIS remains competitive. The MTEP16 backtest supplies only weak-form evidence that BiLo-NSGA selections correlate with real outcomes.

7.1. Evidence-Driven Extensions

The results identify five empirical extensions that would test the framework beyond its current proxy setting.

1. *Proxy review task without expert labels* (Limitation 1). Next step: recruit a small panel of utility planning engineers to independently rank a stratified subset of the 120 candidates, and report the rank correlation between expert priorities and the benchmark's derived attributes; this converts the proxy claim into a measurable calibration statement.
2. *Costs not monetarily calibrated* (Limitation 2). Next step: fit the synthetic cost units against published per-category investment figures – the MISO MTEP cost estimates already cached for the Section 6.5 backtest are a natural anchor – and re-run the flagship budget-constrained scenario with calibrated coefficients to test whether the method ordering is preserved.
3. *External validity below the expert-label rung* (Limitation 3). Next step: extend the MTEP backtest to later plan cohorts (MTEP17–MTEP19), which multiplies the number of explicitly withdrawn projects beyond the current $n = 19$ and gives the strict-label tests the statistical power they presently lack.
4. *Single benchmark family* (Limitation 4). Next step: instantiate a second candidate pool from an independent public system (NREL-118 or a TAMU synthetic case) through the released derivation pipeline; two scenarios with the main baseline comparison suffice to test whether the forward/substitution asymmetry and the budget-margin pattern generalize.
5. *No load-flow verification* (Limitation 5). Next step: run a pandapower AC load-flow check on the compromise portfolios of the two reliability-oriented scenarios and report electrical-violation rates, upgrading portfolio feasibility from budgetary to physical.

The ablation and sensitivity evidence motivates a method redesign. Future atomic substitution should test bounded look-ahead or dominance-based acceptance (Section 6.3), and any depth change should be registered before rerunning the primary family. The completed penalty scan is null because proposals are feasible; future acceptance-rule work should therefore vary objective aggregation rather than the inactive violation coefficient. The small bonus effects provide no basis for retaining 1.06 as an accuracy-tuned value.

8. Limitations

We state the boundaries of this study explicitly.

1. **Proxy review task without expert labels.** The 120 candidates and their attributes are derived from public grid statistics and reliability-report metadata through published rules; no expert-labeled review outcomes exist in the benchmark. All claims are therefore claims about *algorithmic performance on a reproducible public proxy*, not about

- validated real-world review quality. Expert-labeled subsets (with rank-correlation analysis against the proxy) are the direct remedy.
2. **Costs are not monetarily calibrated.** Candidate costs are synthetic cost units generated from network aggregates. Calibrating coefficients against published utility investment figures is required before any engineering-economic claim.
 3. **External checks do not reach the expert-label rung.** The NERC rule backtest retains kind-level construct overlap with pool generation. MTEP16 outcome snapshots were not used in fitting, but decision-time appendix status is prognostic and the project-level tests do not preserve portfolio dependence or a multiplicity family. The approved-plan pool also has an approximately 98% strict-label build rate and almost no renewable projects. These checks are descriptive consistency evidence.
 4. **Single benchmark family.** All eight scenarios instantiate one 120-candidate pool from one RTS-GMLC + SimBench + NERC derivation. A second pool from independent public systems would test whether the forward/substitution asymmetry and the budget-margin pattern generalize.
 5. **No load-flow verification.** Portfolio feasibility is budgetary, not electrical; a pandapower-based load-flow check of selected portfolios is a natural robustness addition.
 6. **The direct local-search comparison is still bounded.** Under both the exact 3200-unit and 0.20-s protocols, BiLo-NSGA significantly exceeds the disclosed add/delete/swap Pareto Local Search in all eight scenarios. However, this is one implementation and does not cover the full family of memetic, large-neighborhood, or exact knapsack algorithms. No universal local-search superiority claim is made.
 7. **The dependency label is only a heuristic search feature.** The 1.06 multiplier changes insertion and replacement proposal order when a group label is already open. No joint benefit, precedence relation, or co-selection constraint is operationalized. The matched sensitivity changes pooled mean hypervolume by only -0.15% at 1.00 and +0.10% at 1.12, descriptively; it does not establish an accuracy gain at 1.06 or dependency synergy.
 8. **Neither experimental record supports exact event replay.** The legacy protocol does not capture every implementation choice needed for exact reconstruction. The matched supplement records its method and accounting choices, but neither archive serializes population states or event payloads. The artifact manifest and file-lineage details are retained in the supplementary record rather than the scientific narrative.
 9. **The matched implementations are stage-local.** The recorded legacy optimizer could not be executed in the matched environment, so the new study uses the fully specified constrained NSGA-II disclosed in Section 5.4. The matched BiLo-NSGA runner also differs from the legacy implementation in normalization-bound timing and denominator floor. Results are valid for these disclosed implementations; equivalence to the earlier implementations is not claimed. Equal-time results are additionally machine-specific.
 10. **The local scan is not an interaction study.** Depth changes the forward and backward proposal caps jointly, and depth, penalty, and bonus are varied one factor at a time. Single-cap effects and interactions are therefore unidentified.

9. Conclusions

BiLo-NSGA defines budget-aware local moves in project-review terms: insertion under slack, atomic delete–insert substitution, a heuristic group-label bonus, deterministic feasibility recovery, and run-level counting of accepted moves and repair drops. The new

effort-controlled evidence does not support superiority over NSGA-II. At exactly 3200 combined population-objective/local-proposal units, pooled feasible-front hypervolume is 0.16277 for the metered stage-local BiLo-NSGA and 0.17213 for the fully specified stage-local NSGA-II. All eight mean differences are negative; four are Holm-significant losses and none is a significant win. The earlier 1.12% fixed-generation margin at 2.74 times the runtime is retained only as an unmatched quality–compute tradeoff.

At equal 0.20-s search deadlines, the NSGA-II comparison is scenario-dependent: three significant wins, two significant losses, and three unresolved contrasts. BiLo-NSGA exceeds the bounded Pareto Local Search in all eight scenarios under both matched protocols, but no family-wide local-search claim follows. Every final population in the 1440 comparison runs is budget-feasible. Six bound/clipping/reference schemes preserve the pooled NSGA-II–BiLo–PLS ordering while changing scales and gaps; the joint (2, 2) proposal-cap setting is descriptively 4.29% above the registered (8, 4) setting, the group-bonus effects are below 0.15% in magnitude, and the feasible-proposal penalty scan is exactly null.

The earlier component, NERC, and MTEP16 analyses retain their narrower descriptive scope. Forward insertion resolves only in dependency-constrained review, local-move explainability, and project-pool scalability under the declared legacy family; atomic substitution and standalone deletion remain unresolved in all eight scenarios. The study supports BiLo-NSGA as a feasible project-vocabulary search framework on the public proxy benchmark and identifies a quality–effort boundary for its registered local allocation. It does not establish NSGA-II superiority, universal forward benefit, substitution benefit, dependency synergy, audit completeness, lineage, replay, a recommendation path, or a preferred operational configuration. Expert labels, calibrated costs, a second benchmark family, broader local-search comparisons, and load-flow verification remain necessary to evaluate practical review effectiveness.

Author Contributions

[AUTHOR INPUT REQUIRED: assign the CRediT roles to Yubin Lin, Jingbo Zhang, Xiaoyu Huang, Dishan Yang, and Jiyu Li, and obtain approval from every author.] All authors have read and agreed to the published version of the manuscript.

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Data Availability Statement

All data used in this study are public. Candidate projects are derived from RTS-GMLC (<https://github.com/GridMod/RTS-GMLC>), SimBench (<https://simbench.de>), and meta-data from public NERC reliability reports (<https://www.nerc.com>). MISO MTEP16 records are available through the MISO transmission-planning archive (<https://www.misoenergy.org/planning>). The supplementary review package contains the legacy 4320-run archive, the matched-effort run records and sensitivity analyses, corrected inference, validation checks,

and the artifact manifest used to regenerate the reported tables and figures. Neither experimental archive serializes event payloads or population states. A persistent public archive can be supplied before publication, subject to third-party redistribution terms. The TRACE-MOEA companion study shares the candidate generator, source corpora, common benchmark/evaluation utilities, and public-record backtest infrastructure. Problem objectives, method configurations, executions, run outputs, selected portfolios, and reported comparisons are paper-specific.

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Conflicts of Interest

The authors declare no conflicts of interest.

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